Development of an LQR-Based Control Algorithm for Quadcopter

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***Abstract*— The increasing use of unmanned aerial vehicles (UAVs) necessitates the development of more sophisticated control algorithms to enhance the performance of rotary-wing UAVs. This article discusses the implementation of a linear quadratic regulator (LQR) controller as a control technique for the flight controller of the Qdrone1, developed by Quanser. While LQR analysis is well-known, our approach distinguishes itself through the integration of an advanced six-camera system to establish an exceptionally accurate reference point for the drone. This innovative use of the OptiTrack system for precise position tracking feeds into the control algorithm, significantly enhancing the accuracy and stability of the UAV. The LQR controller is evaluated in real-time using the MATLAB/Simulink environment, including control loops for altitude, position, and attitude. The primary objective is to develop an optimal controller for effectively regulating the drone's motion while maintaining the utmost stability and accuracy. Using its dynamic model, a MATLAB Simulink algorithm is developed to control the quadrotor. This work further demonstrates the creation and verification of a simulation model for Qdrone1. The correctness of the model is evaluated by comparing its simulated outcomes with data obtained from flight testing. The flight data and simulation results show the model's reliability. The main goal of this research is to develop a precise dynamic model and LQR controller for Qdrone1, allowing for the comprehensive evaluation of this specific UAV configuration.**

Keywords—Quadcopter, Flight controller, Model-based design, Linear quadratic regulator (LQR) controller, Modelling.

I. Introduction

Unmanned aerial vehicles (UAVs) have been significantly contributing to many civilian applications over the last several decades. Specifically, rotary-wing UAVs have been used in many applications, and shortly, corporations intend to expand their utilization further. Agricultural operations extensively use UAVs for tasks such as field mapping, plant stress detection, and chemical spraying. In some circumstances, drones are already ensuring optimal performance from outdated equipment, resulting in a significant decrease in operating times [1]. Recently UAVs have been used for aerial surveillance and surveying purposes. These drones may be fitted with remote sensing cameras to identify water sources in arid regions or locate water leaks in underground pipes [2]. Additionally, some are used to gather data from areas impacted by natural calamities. Within a decade UAVs might be extensively used for delivery purposes, taking advantage of the vertical dimension [3]. This would result in a significant reduction in delivery times and less congested streets.

Flight controllers for UAVs play a critical role in ensuring optimal flying performance and exceptional stability for these aircraft [4]. Specifically, rotary-wing UAVs possess inherent static and dynamic instability, necessitating the implementation of precise control algorithms to ensure stable flight and execute maneuvers. Many flight controllers in commercial rotary-wing UAVs use a proportional-integrative-derivative (PID) control algorithm, which can be readily adjusted and executed on a microcontroller [5]. PID control is effective in ensuring flight stability. However, the growing demand for UAVs has prompted several academics to devise and evaluate more sophisticated control systems. Capello et al. introduced an L1 adaptive controller as the autopilot for the inner loop of a fixed-wing UAV [6]. Kyaw and Gavrilov used the L1 adaptive controller to manage the attitude control of a quadcopter. They also presented a straightforward method for designing the controller. Model predictive control (MPC) has mostly been evaluated using simulation models [7]. This control technique demonstrates the benefit of not relying on model parameters, making it particularly efficient in addressing nonlinearities and uncertainties in the model. Islam et al. conducted a successful simulation-based experiment to investigate the effectiveness of a linear Model Predictive Control (MPC) system for trajectory control of a quadcopter UAV [8]. Suhail highlighted the adoption of a unique control strategy termed active disturbance response (ADR) [9] because of the quadcopter's susceptibility to disturbances and the significant nonlinear effects it experiences. The results presented in the research show that the ADR has a superior capacity to respond quickly to disturbances compared to a PID controller [10]. Additionally, neural network (NN) methods have been suggested for constructing the flight controller's inner loop [11]. A simulation tool was developed to create a model-based controller using a radial basis function neural network trained with the minimum resource allocating

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network (MRAN) algorithm [12], offering an accurate representation of quadcopter attitude dynamics [13].

This study presents a novel approach by integrating an advanced six-camera OptiTrack system for exceptionally accurate reference point establishment and precise position tracking for the Qdrone1. This innovative integration significantly enhances the LQR controller's accuracy and stability. The LQR-based controller with integrative action was designed using MATLAB/Simulink software, tailored to Qdrone's specific characteristics. The controller's responses were tested with control inputs and reference trajectories, demonstrating superior performance. This unique integration and real-time evaluation in a MATLAB/Simulink environment set this work apart from existing control techniques.

II. Mathematical Model of Quadcopter

This section outlines the creation of a mathematical model for the quadcopter that considers non-linear factors. Before constructing the model, the following assumptions are made:

- The quadcopter is classified as a rigid body, which implies that any vibrations or deformations that occur during flight are disregarded.
- The motor's thrust is unaffected by external influences such as variations in air density, temperature, and altitude [15].
- An Inertial Reference Frame $\vec{E}:\{\vec{e_1},\vec{e_2},\vec{e_3}\}$ Is a coordinate system anchored to the Earth that usually places its starting point at ground level.
- The Body Reference Frame$\vec{B}:\{\vec{b_1},\vec{b_2},\vec{b_3}\}$ Is defined with its origin located at the center of gravity (COG) of the quadrotor.
- The origin of the Horizon Reference Frame $\vec{H}:\{0,0,\vec{h_3}\}$ is the quadrotor's mass center, and its axes are oriented in the direction of the horizon, or the nearby vertical and horizontal planes shown in Fig 1.

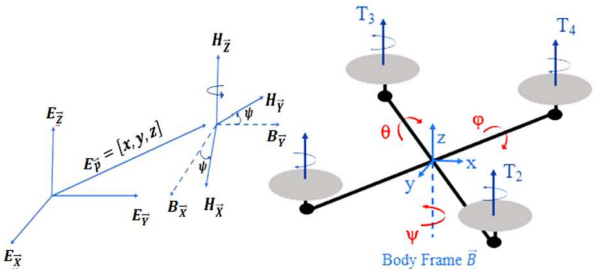

Fig. 1. North-east-down (NED) reference system [12].

A. Nonlinear Dynamic Equation of Quadcopter using Euler's equation

The motion of the quadcopter is described in both the body and inertial frames. In the inertial frame, the quadcopter's position and velocity are represented as $\boldsymbol{x}=(x, y, z)^T$ and $\boldsymbol{V} = [\dot{x},\dot{y},\dot{z}]^T$ Respectively. In the body frame, the velocity is represented as $\boldsymbol{V_B} = [u,v,w]^T$. we define the roll, pitch, and yaw angles in the body frame as $\omega = (\phi, \theta, \psi)^T$, The vector. $\Omega = [\dot{\phi},\dot{\theta},\dot{\psi}]^T$ represents the angular rates [14]. It is crucial to comprehend that the angular velocity vector Ω is not equal to the angular velocity. $\dot{\omega}$. The angular velocity is a vector that specifies the rotational direction, whereas $\dot{\theta}$ represents the rate of change of the yaw, pitch, and roll in an inertial frame the relation is given as: [16].

$$\Omega = \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 0 & -s\theta \\ 0 & c\phi & s\phi c\theta \\ 0 & -s\phi & c\phi c\theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} \tag{1}$$

$$\Omega = M\dot{\omega}\,, M \in \mathbb{R}^{3\times 3}$$

The quadrotor's location and rotation vector can be represented as $\boldsymbol{q} = [X\ \omega]^T$. The vector is defined as $[x\ y\ z\ \phi\ \theta\ \psi]^T$. The linear and angular velocities are represented by the vector $\mathbf{V} = [V\ \Omega]^T$, where V represents the linear velocities in the x, y, and z directions, and Ω represents the angular velocities around the p, q, and r axes $\boldsymbol{V} = [V\ \Omega]^T = [v_x\ v_y\ v_z\ p\ q\ r]^T$. The translational motion is determined by the positional state variables. $P_{\vec{E}}$ in the inertial frame and the positional state variables $P_{\vec{B}}$. In the body frame. R is Rotation matrices in eq (3):

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}_{\vec{E}} = R \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{\vec{B}} \tag{2}$$

$$R = \begin{bmatrix} c\theta c\psi & s\phi s\theta c\psi - c\phi s\psi & c\phi s\theta c\psi + s\phi s\psi \\ c\theta s\psi & s\phi s\theta s\psi + c\phi s\psi & c\phi s\theta s\psi - s\phi c\psi \\ -s\theta & s\phi c\theta & c\phi c\theta \end{bmatrix} \tag{3}$$

In equation (3) sin(τ) is replaced with sτ and cos(τ) is replaced with cτ where $\tau = \theta, \phi, \psi$.

The quadrotor's translational and rotational kinetic energy is determined by the following equations [16]:

$$T_{\text{trans}} = \frac{1}{2} m \dot{W}^T \dot{W}$$

$$T_{\text{rot}} = \frac{1}{2} \dot{\Omega}^T J \dot{\Omega}$$

and $J = \text{diag}\{J_x \quad J_y \quad J_x\} \in \mathbb{R}^{3\times 3}$ matrix represents the inertia of the quadrotor in the body-fixed frame, which is a result of its symmetry. The quadrotor's Euler-Lagrange equation is provided [17]:

$$\frac{d}{dt}\left(\frac{\delta L}{\delta \dot{q}}\right) - \frac{\delta L}{\delta q} = \begin{bmatrix} F \\ \tau \end{bmatrix} \tag{4}$$

F represents the external force applied to the quadrotor. τ represents the generalized moments around the UAV. The equation for force is given by $F = F_X + F_t$ and $\tau = [\tau_\phi \quad \tau_\theta \quad \tau_\psi]^T$ where F_X is the force of translational movement created by propellers and F_t is the drag forces acting on the quadrotor along $\{b_1, b_2, b_3\}$ vectors, $f = \sum_{t=1}^{4} f_1$ and $F_X = f R_{e_3}$.

$$F_X = f \begin{bmatrix} c\phi s\theta c\psi + s\phi s\psi \\ c\phi s\theta s\psi - s\phi c\psi \\ c\phi c\theta \end{bmatrix} \tag{5}$$

$$F_t = \begin{bmatrix} -K_{fx} & 0 & 0 \\ 0 & -K_{fy} & 0 \\ 0 & 0 & -K_{fz} \end{bmatrix} \dot{X} \tag{6}$$

where K_{fx}, K_{fy}, K_{fz} are drag coefficients, now putting F in equation (5).

$$m \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = f \begin{bmatrix} c\phi s\theta c\psi + s\phi s\psi \\ c\phi s\theta s\psi - s\phi c\psi \\ c\phi c\theta \end{bmatrix} + \begin{bmatrix} -K_{fx} & 0 & 0 \\ 0 & -K_{fy} & 0 \\ 0 & 0 & -K_{fz} \end{bmatrix} \dot{X} - mg \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{7}$$

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$$m\ddot{x} = (c\phi s\theta c\psi + s\phi s\psi)U_1 - K_{fx}\dot{x} \quad (8)$$

$$m\ddot{y} = (c\phi s\theta s\psi - s\phi c\psi)U_1 - K_{fy}\dot{y} \quad (9)$$

$$m\ddot{z} = (c\phi c\theta)U_1 - mg - K_{fz}\dot{z} \quad (10)$$

Whereas $\ddot{x}, \ddot{y}$ and $\ddot{z}$ represent linear accelerations in the horizon frame. The Euler equation, $M = J\dot{\Omega} + \Omega \times J\Omega$ yields the following three angular motion equations [12]:

$$J\dot{\Omega} = \Omega \times J\Omega + \sum_{i=1}^{4} J_r(\Omega \times e_3)\Omega_i + \tau \quad (11)$$

$$\begin{bmatrix} j_x & 0 & 0 \\ 0 & j_y & 0 \\ 0 & 0 & j \end{bmatrix} \begin{bmatrix} \ddot{\phi} \\ \ddot{\theta} \\ \ddot{\psi} \end{bmatrix} = \begin{bmatrix} \dot{\theta}\dot{\psi}(j_z - j_y) \\ \dot{\phi}\dot{\psi}(j_x - j_z) \\ \dot{\theta}\dot{\phi}(j_y - j_x) \end{bmatrix} + \begin{bmatrix} J_r\dot{\theta}\Omega_r \\ -J_r\dot{\phi}\Omega_r \\ 0 \end{bmatrix} + \begin{bmatrix} \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} \quad (12)$$

$$\ddot{\phi} = \dot{\theta}\dot{\psi}k_{1_\phi} - k_{2_\phi}\dot{\theta}\Omega_r + l_\phi U_2 \quad (13)$$

$$\ddot{\theta} = \dot{\phi}\dot{\psi}k_{1_\theta} - k_{2_\theta}\dot{\psi}\Omega_r + l_\theta U_3 \quad (14)$$

$$\ddot{\psi} = \dot{\phi}\dot{\theta}k_{1_\psi} + U_4 \quad (15)$$

Where, $k_{1_\phi} = (j_y - j_z)/j_x$, $k_{2_\phi} = \frac{j_r}{j_x}$, $k_{1_\theta} = (j_z - j_x)/j_y$, $k_{1_\theta} = j_r/j_y$, $k_{1_\psi} = (j_x - j_y)/j_z$, $l_\phi = l/j_x$, $l_\theta = l/j_y$ and $l_\psi = l/j_z$ with l representing the length, Ω_r denotes rotor inertia $\Omega_r = \omega_2 + \omega_4 - \omega_1 - \omega_3$.

III. Linearization Of the Quadcopter Model

When the sets of equations specified in Equations (8)-(10) and (13)-(15) are examined, it contains nonlinear expressions. These equation sets need to be linearized since this research will use linear control approaches. The equilibrium point for linearization was identified as the quadcopter's hover position, with the position vector represented as $p = [x\ y\ z]^T$ and the velocity vector as $n = [0\ 0\ 0]^T$. Consider the equation to be linearized [17].

$$Y''(t) = f(Y(t), Y'(t), U(t), U'(t))$$

The procedure for linearizing this equation is as follows:

- We calculate the partial derivatives with respect to its different variables and we evaluate these derivatives at the equilibrium point considered.
- We write the expansion limited to order 1 of the function f, neglecting the higher order terms to reduce the function f to its linear part only.
- We write from our initial equation, the equation of the resulting behavior, namely:

$$Y''(t) = \frac{\partial f}{\partial Y}\Big|_{Y0,0,U0,0} * (Y(t)-Y_0) + \frac{\partial f}{\partial y'}\Big|_{Y0,0,U0,0} * (Y'(t)-0) + \frac{\partial f}{\partial U}\Big|_{Y0,0,U0,0} * (U(t)-U_0) + \frac{\partial f}{\partial U'}\Big|_{Y0,0,U0,0} * (U'(t)-0). \quad (16)$$

In the small angle's approximation, the cosine values are an approximation. By applying the first-order Taylor expansion to the Dynamic Equation at the equilibrium point, we get the linear model of the quadrotor.

$$\ddot{x} = g\theta \quad (17)$$

$$\ddot{y} = -\phi g \quad (18)$$

$$m\ddot{z} = T \quad (19)$$

$$\ddot{\phi} = \frac{U_\phi}{j_x} \quad (20)$$

$$\ddot{\theta} = \frac{U_\theta}{j_y} \quad (21)$$

$$\ddot{\psi} = \frac{U_\psi}{j_z} \quad (22)$$

IV. Development Of LQR Algorithm For Quadcopter

The linear quadratic regulator is an optimum linear control approach that conducts optimization over system dynamics using feedback from all states. First it is necessary to model the system that is to be controlled in the following state space equations [18]:

$$\dot{x} = Ax + Bu \quad (23)$$

$$y = Cx + Du \quad (24)$$

$$u(t) = -Kx(t) \quad (25)$$

K is optimized using the cost function in LQR controller design:

$$J = \int_0^\infty (x^T Q x + u^T R u)dt \quad (26)$$

In Equation (25), Q represents the weight matrix and R represents the control matrix. The determination of the optimum coefficient is performed in the following manner [19].

The state variables and inputs for height in this system are as follows:

$$x_z = [z\ \dot{z}]^T, u_z = T - mg \quad (27)$$

The state-space representation is given by the following equation:

$$\dot{x_z} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x_z + \begin{bmatrix} 0 \\ 1/m \end{bmatrix} u_z \quad (28)$$

$$y_z = [1\ \ 0]x_z \quad (29)$$

Position Updates in the x and y coordinates of the quadcopter also impact the rotation around the longitudinal and lateral axes in the reference frame of the quadcopter. The modifications may be expressed in the following manner for the transnational Equation:

$$\left.\begin{aligned} \ddot{x}_d &= \frac{T}{m}(\cos\phi\sin\theta) \\ \ddot{y}_d &= \frac{T}{m}(-\sin\phi) \\ \ddot{z}_d &= \frac{T}{m}(\cos\phi\cos\theta) \end{aligned}\right\} \begin{aligned} \theta_d &= \operatorname{atan}\left[\frac{\ddot{x}_d}{\ddot{z}_d + g}\right] \\ \phi_d &= \operatorname{atan}\left[\frac{-\ddot{y}_d\cos\theta}{\ddot{z}_d + g}\right] \end{aligned} \quad (30)$$

State-space representation for reference position control in the X-Y axis is:

$$x_{Pos} = [\,^B x_I\ \ddot{x}\ \ ^B y_I\ \ddot{y}]^T, u_x = \theta, u_y = \varphi \quad (31)$$

$$y_{Pos} = [x\ y]^T$$

$$\dot{x}_{Pos} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ \dot{x} \\ \dot{y} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & g \\ -g & 0 \end{bmatrix} u_{Pos} \quad (32)$$

$$y_{Pos} = [1\ 1]x_z \quad (33)$$

The state variable and inputs for the altitude Controller are expressed as:

$$\boldsymbol{x}_{at} = [\phi_d\ \dot{\phi}\ \ \theta_d\ \dot{\theta}\ \ \psi\ \dot{\psi}]^T, u_\theta = \tau_\theta \quad (34)$$

$$, u_\phi = \tau_\phi, u_\psi = \tau_\psi$$

$$y_{at} = [\phi_d\ \theta_d\ \psi\ ]^T$$

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$$\dot{x}_{at} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{1}{j_x} & 0 & 0 \\ 0 & \frac{1}{j_y} & 0 \\ 0 & 0 & \frac{1}{j_z} \end{bmatrix} u_{at} \tag{35}$$

$$y_{at} = [1\ 1]x_z \tag{36}$$

The Q and R matrices for each controller were adjusted according to these linear models. This approach offers a more straightforward method for the tuning process since the linear equations are largely independent of each other. This means that the adjustment of the specified matrices may be made without worrying about how it would impact the stability of other states, unless in the case of a cascaded situation.

Qz = [0.03 0; 0 0.05].
Rz = .0002.
Qpos=diag ([0.1 0.1 0.001 0.001]).
Rpos=diag ([1 1]).
Qatt=diag ([20 17 .15 .05 .05 .09]).
Ratt=diag ([1 1 1]).

In this procedure, the most optimal gain matrix K is determined using MATLAB for the linear model and then validated on the non-linear model in Simulink.

The controller is designed according to the altitude, position, and attitude loop structure, as shown in Fig 2.

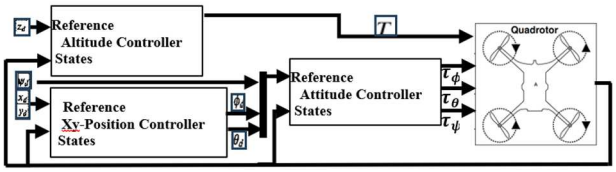

Fig. 2. The altitude, position, and attitude loop structure used for the LQR controller.

The LQR controller requires determining the Q and R matrix, which can be done through trial and error, and 18 coefficients, which govern the quadcopter's x, y, and z locations, without using time-consuming optimization procedures like GA.

V. Experimental Setup

A. Hardware

The experiments are carried out inside a 5.5x3.5 square meter area that is surrounded by a safety net shown in Fig 4a. As seen in Fig 4b, a total of six cameras are positioned at the corners and edges of the workplace. These cameras are linked to a computer via wires. OptiTrack utilizes cameras to ascertain the precise position and orientation of the UAV. The current UAV in use is a Qdrone1, which is produced by Quanser, Canada. As per the manufacturer's prescribed plan, five markers were affixed on the Qdrone1, as shown in Fig 3.

B. Real-time Communication and Localization

The vehicles and computers communicate via a router and wireless Wi-Fi connection, providing a unique IP address for the computer, drone, and robot. MATLAB Simulink and QUARC blocks facilitate vehicle management.

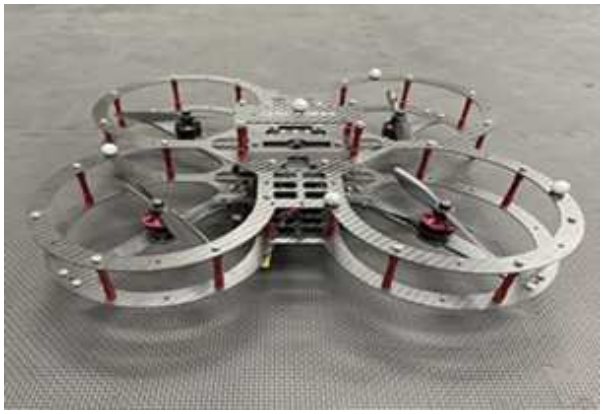

Fig. 3. QUANSER Qdrone1.

C. Experimental procedure

Fig 5 illustrates the steps of the operation technique involving the OptiTrack system detecting the quadcopter's location, which is sent to the Mission Server. The server computers and transmits the position to the Stabilizer. The drone tracks the specified location.Qdrone1 lands after receiving instructions.

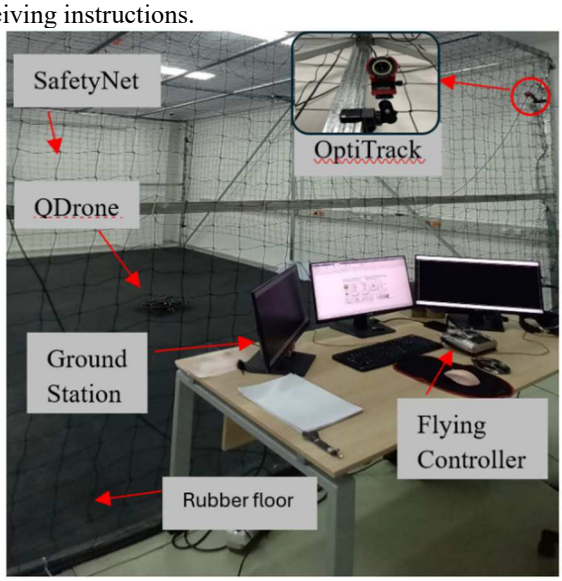

Fig. 4(a). Enclosed workspace with a safety net and rubber floor.

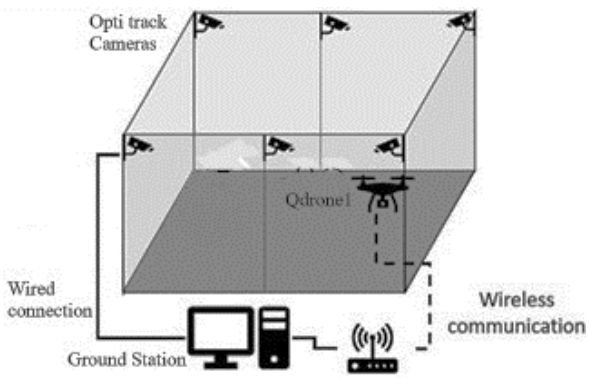

Fig. 4(b). Communication scheme[12].

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D. *MATLAB Simulink*

A Wi-Fi connection and router enable communication between the onboard CPU of Qdrone1 and the ground station (PC). The control algorithm and mission server are implemented using MATLAB. Table 1 presents the precise technical details of Qdrone1. When comparing Qdrone1 dynamics to other quadrotors, the key difference in modeling is that the lengths of $LRoll$ and $Lpitch$ are not equal.

TABLE I. Description of quadcopter parameter.

Symbol	Description	Value	Units
Jx	Roll Moment of Inertia	0.01	kg*m^2
Jy	Pitch Moment of Inertia	0.082	kg*m^2
Jz	Yaw Moment of Inertia	0.015	kg*m^2
b	Maximum commanded thrust force	5.11	N
d	Normalized yaw torque constant	0.048	Nm
M	Total mass of the quadcopter	1.121	kg
g	Gravitational constant	9.81	m/s^2
L_$Roll$	Roll motor-to-motor distance	0.213	m
L_P-$itch$	Pitch motor-to-motor distance	0.175	m

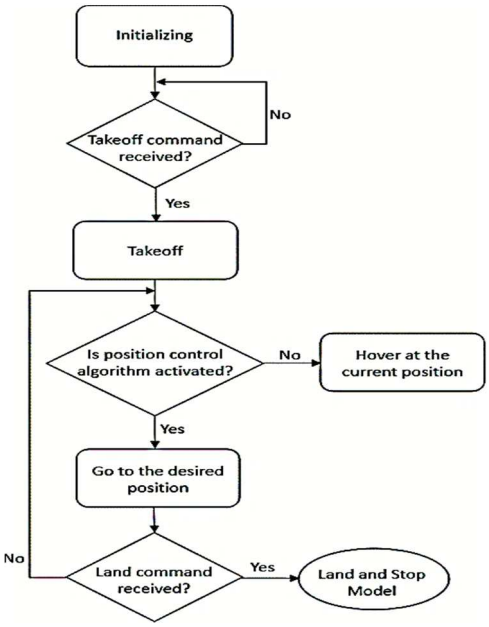

Fig. 5. Operation procedure flowchart [12].

VI. Result And Validation

The drone's position tracking experiments revealed that a Linear Quadratic Regulator (LQR) controller provided the best stable control in the x, y, and yaw axes, as measured by root mean square error (RMSE) and peak-to-peak (P2P) data shown in Fig 6(a)-6(d). The LQR controller strikes a balance between control effort and system performance by minimizing a cost function that penalizes departures from the intended trajectory and excessive control inputs. This provides accurate tracking performance with minimum variance.

A. *Stability of Each Axis:*

The drone's position tracking tests showed that an over LQR controller achieved the most stable behavior for x, y, and yaw position control. This controller, designed to balance control effort and system performance, minimizes deviations from the target trajectory and excessive control inputs. The quadrotor's precise and effective tracking performance ensures the drone accurately follows the target route with little deviation shown in Table II, demonstrating the effectiveness of this controller in drone position tracking.

Yaw Axis refers to the horizontal axis around which an object rotates or moves. The LQR controller stabilizes the yaw axis, controlling rotational movement around the vertical axis, by making differential thrust changes.

TABLE II. Numerical Results

Metric	**x-axis (m)**	**y-axis (m)**	**z-axis (m)**	**Yaw (rad)**
RMSE	**0.04**	**0.05**	**0.001**	**0.01**
(P2P)	**0.10**	**0.12**	**0.01**	**0.02**
Total Accuracy (%)	**98**	**97**	**98**	**97**
Accumulated Error	**0.07**	**0.08**	**0.001**	**0.005**

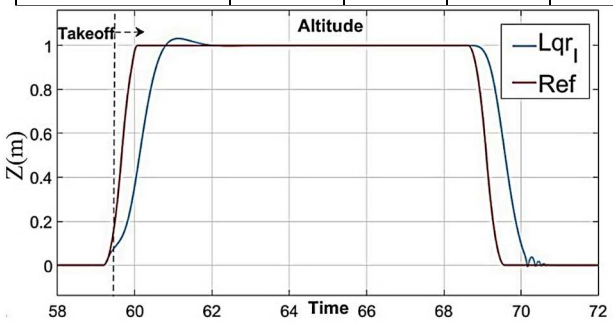

Fig.6(a). Z Position vs. time (Position control test, LQR)

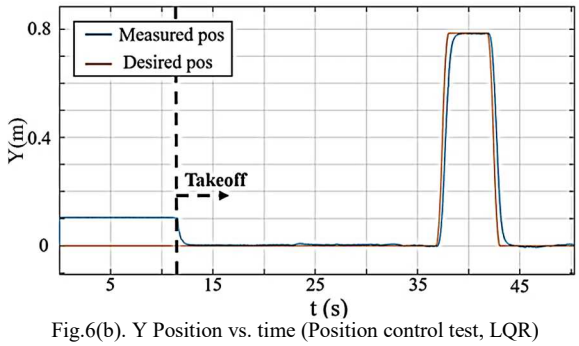

Fig.6(b). Y Position vs. time (Position control test, LQR)

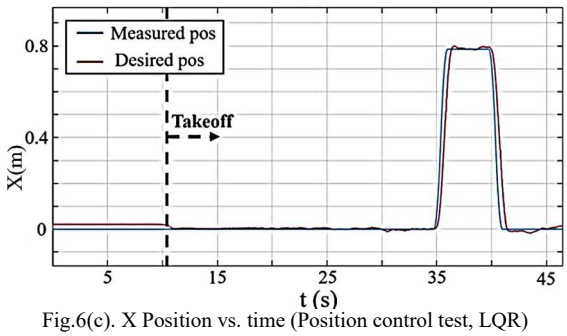

Fig.6(c). X Position vs. time (Position control test, LQR)

B. *Controller Action*

The LQR controller iteratively computes optimal control inputs based on the quadrotor's current state and the planned trajectory, considering variables like location, velocity, direction, and angular velocity. It solves the Riccati equation and applies the feedback gain matrix to address system discrepancies. The LQR controller minimizes a quadratic cost function to determine the optimum control law,

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balancing performance and energy efficiency. It dynamically adjusts motor thrusts to realign the quadrotor with the target route, minimizing overshooting shown in Fig 7(a),7(b).

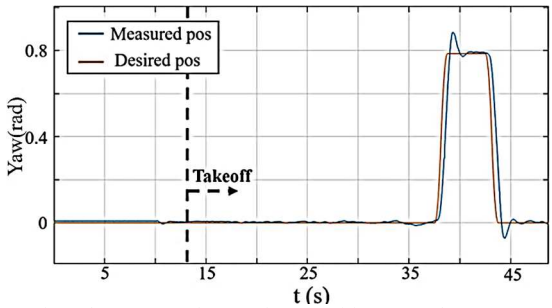

Fig.6(d). Yaw rotation vs. time (Position control test, LQR)

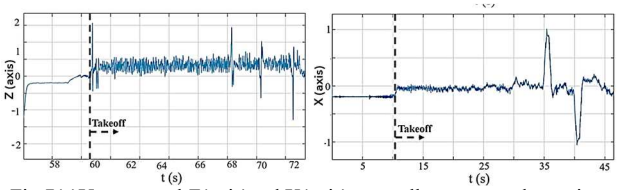

Fig.7(a)Unsaturated Z(axis)and X(axis) controller commands vs. time (LQR)

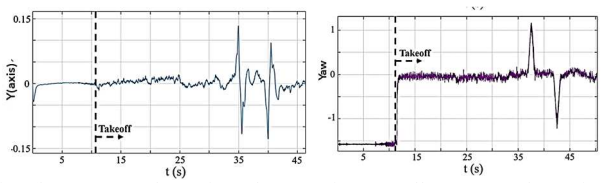

Fig.7(b)Unsaturated Y(axis)and Yaw(axis) controller commands vs. time (LQR)

VII. Conclusion And Future Work

In this work, the construction and verification of a simulation model for the Qdrone1 quadcopter have been shown by comparing the results of the simulation with actual data collected in the real world. Although the system dynamics are extremely nonlinear and interrelated, the model is nevertheless useful for developing a control algorithm utilizing a model-based approach. The experimental approach for determining physical parameters is thoroughly explained, demonstrating that the quadcopter's dynamics can be precisely calculated using Newton's and Euler's equations, even without including wind disturbances and other nonlinear factors. Although the model may show discrepancies when instructions change rapidly and significantly, it consistently and properly predicts the time it takes for something to rise and effectively generates the necessary control settings.

Future endeavors will prioritize the incorporation of a Disturbance Observer (DOB) to augment the precision of the model by considering external disturbances such as wind and unexpected changes in the payload. In addition, the study will focus on enhancing control algorithms to maximize the quadcopter's maneuverability and stability in different situations, guaranteeing consistent and dependable performance in real-world uses. This work establishes a strong basis for future progress in UAV operations, emphasizing the effectiveness of the LQR controller in preserving stability and accuracy, thereby improving the resilience and reliability of quadcopter control algorithms.

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